

A Sigma-Compact AM-Space Subcase for Nonlinear Order Isomorphisms

Automatic Math Discovery Smoke Test

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Source and Extracted Question

The source is Denny H. Leung and Wee-Kee Tang, *Persistence of Banach lattices under nonlinear order isomorphisms*, arXiv:1507.02759. After proving that a Banach lattice order isomorphic to $C(K)$ is lattice isomorphic to $C(K)$, the authors write:

We do not know if a Banach lattice that is order isomorphic to an AM -space must be lattice isomorphic to an AM -space.

$C(K)$ and $C(L)$ are nonlinearly order isomorphic. By [1, Proposition 3], K and L are homeomorphic. Thus $C(K)$ and $C(L)$ are lattice isomorphic. Since E is lattice isomorphic to $C(L)$, the proof is complete. $\square$

We do not know if a Banach lattice that is order isomorphic to an AM -space must be lattice isomorphic to an AM -space. In this direction, there is a useful characterization of AM -spaces due to Cartwright and Lotz; see [2] and [5, Theorem 2.1.12]. A subset A in an ordered vector space E is *order bounded* if there are $u, v \in E$ such that $u < x < v$ for all $x \in A$. A sequence

The source proves a partial result when the target is a closed ideal in $C(K)$ and $C(K)$ is separable. This packet records a broader positive subcase.

Result

Theorem 1. *Let L be a locally compact Hausdorff sigma-compact space. If a Banach lattice E is order isomorphic to $C_0(L)$, then E is lattice isomorphic to an AM -space.*

Idea of the Proof

The Cartwright–Lotz criterion says that a Banach lattice is lattice isomorphic to an AM -space exactly when every disjoint norm-null sequence is order bounded in the bidual. We prove something slightly stronger: each such sequence in E is already order bounded in E .

Let $T : E \rightarrow C_0(L)$ be an order isomorphism with $T0 = 0$. If (x_n) is disjoint and norm-null in E , then

$$f_n = T|x_n|$$

is a disjoint positive sequence in $C_0(L)$. The summability of a fast subsequence of (x_n) prevents (f_n) from being unbounded, and also prevents the family (f_n) from having a uniformly positive tail escaping to infinity. Sigma-compactness then lets us build a single positive $h \in C_0(L)$ with $f_n \leq h$ for all n . Therefore $|x_n| \leq T^{-1}h$ for all n .

Envelope Lemma

Lemma 1. *Let L be locally compact Hausdorff and sigma-compact. Let $(f_n) \subset C_0(L)_+$ be bounded. Suppose that for every $\varepsilon > 0$ there is a compact set $K \subset L$ such that*

$$\sup_n \sup_{t \in L \setminus K} f_n(t) \leq \varepsilon.$$

Then there is $h \in C_0(L)_+$ such that $f_n \leq h$ for every n .

Proof. Let $M = \sup_n \|f_n\|_\infty$. Choose compact sets

$$K_1 \subset \text{int } K_2 \subset K_2 \subset \text{int } K_3 \subset \cdots, \quad \bigcup_m K_m = L,$$

such that

$$\sup_n \sup_{L \setminus K_m} f_n \leq 2^{-m} \quad (m \geq 1).$$

By Urysohn's lemma for locally compact Hausdorff spaces, choose $\varphi_m \in C_c(L)$ with $0 \leq \varphi_m \leq 1$, $\varphi_m = 1$ on K_m , and $\text{supp } \varphi_m \subset \text{int } K_{m+1}$. Set

$$h = M\varphi_1 + \sum_{m=1}^{\infty} 2^{-m} \varphi_{m+1}.$$

The series converges uniformly and $h \in C_0(L)_+$.

If $t \in K_1$, then $h(t) \geq M \geq f_n(t)$. If $t \notin K_1$, let $m \geq 2$ be the least index with $t \in K_m$. Then $t \notin K_{m-1}$, so $f_n(t) \leq 2^{-(m-1)}$ for every n , while the term $2^{-(m-1)}\varphi_m$ in the definition of h gives $h(t) \geq 2^{-(m-1)}$. Thus h dominates all f_n . $\square$

Proof of Theorem 1

Proof. Let $T : E \rightarrow C_0(L)$ be an order isomorphism and replace T by $x \mapsto T(x) - T0$, so that $T0 = 0$.

Let (x_n) be a disjoint norm-null sequence in E . Put

$$f_n = T|x_n| \in C_0(L)_+.$$

The sequence (f_n) is disjoint, because T preserves lattice order and therefore preserves finite infima and suprema.

First, (f_n) is norm bounded. If not, pass to a subsequence, still denoted (n_k) , such that

$$\|x_{n_k}\| \leq 2^{-k}, \quad \|f_{n_k}\|_\infty > k.$$

Then $x = \sum_k |x_{n_k}|$ converges in E , and $x \geq |x_{n_k}|$ for all k . Hence $Tx \geq f_{n_k}$ for all k , forcing

$$\|Tx\|_\infty \geq \|f_{n_k}\|_\infty > k$$

for every k , a contradiction.

Next, the family (f_n) is uniformly vanishing at infinity. Suppose not. Then for some $\varepsilon > 0$, for every compact $K \subset L$ and every N there are $n \geq N$ and $t \in L \setminus K$ with $f_n(t) > \varepsilon$. Choose a compact exhaustion (K_k) of L and inductively choose $n_k > n_{k-1}$ and $t_k \in L \setminus K_k$ such that

$$\|x_{n_k}\| \leq 2^{-k}, \quad f_{n_k}(t_k) > \varepsilon.$$

Again $x = \sum_k |x_{n_k}|$ converges in E , so $Tx \geq f_{n_k}$ for all k . Thus $Tx(t_k) > \varepsilon$ for every k . But t_k eventually leaves every compact subset of L , while $Tx \in C_0(L)$, a contradiction.

By Lemma 1, choose $h \in C_0(L)_+$ such that $f_n \leq h$ for all n . Applying T^{-1} , we get

$$|x_n| \leq T^{-1}h \quad (n \in \mathbb{N}).$$

Thus every disjoint norm-null sequence in E is order bounded in E . In particular it is order bounded in E'' . By the Cartwright–Lotz theorem [1], E is lattice isomorphic to an AM -space. $\square$

Novelty and Limitations

The run indexes were searched for 1507.02759, nonlinear order isomorphism, AM -space, and sigma-compact variants; no duplicate packet was found. Bounded web searches for the exact source question and close sigma-compact phrases found the source arXiv page, but no later direct answer to this subcase.

This is not a full solution of the source question. The proof depends on sigma-compactness of the locally compact representation space L , via the single-envelope Lemma 1. The arbitrary non-sigma-compact AM -space case remains open here.

References

- [1] D. I. Cartwright and H. P. Lotz, *Disjunkte Folgen in Banachverbänden*, Archiv der Mathematik **24** (1973), 51–59.
- [2] D. H. Leung and W.-K. Tang, *Persistence of Banach lattices under nonlinear order isomorphisms*, arXiv:1507.02759.